

GH RAISONI COLLEGE • DEPARTMENT OF BBA-CA • SYBBA-CA (SEM III)

AI CHATBOT FOR COLLEGE — REVISED DIAGRAMS

Interaction boundaries, primary actors, and secondary system storage actor

1. Use Case Diagram (With Secondary Database Actor)

✓ Teacher Correction Applied: Added <> Database System (SQLite 3) with solid associations to all data operations.

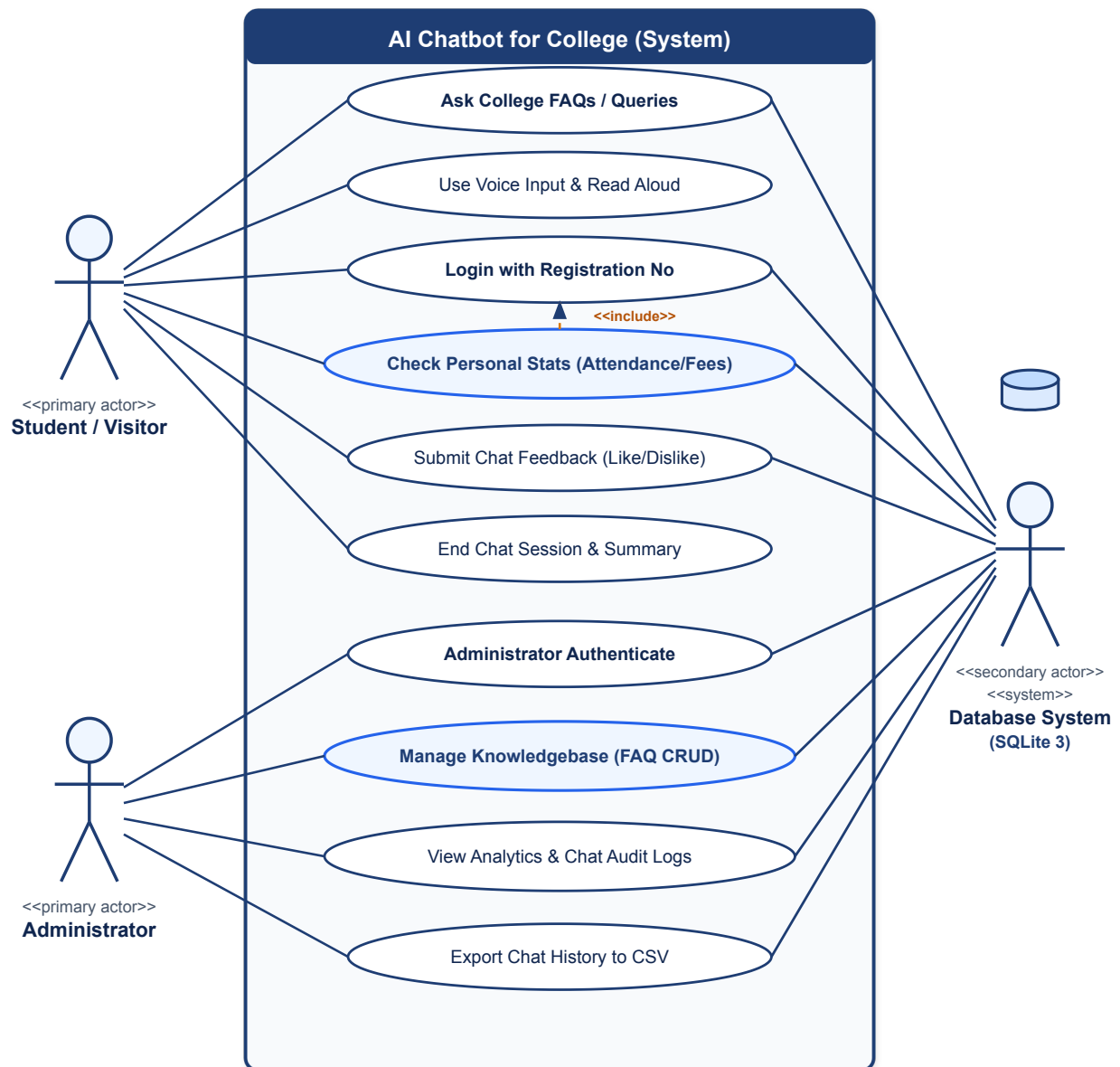


Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and secondary actor (SQLite Database System) with data-tier associations.

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Structural classes, attributes, methods, and solid directed relationships

2. Class Diagram (With 100% Solid Association Arrows)

✓ Teacher Correction Applied: Replaced all dashed lines with 100% solid lines & solid arrowheads; marked multiplicities (1 to 0..*) and role verbs.

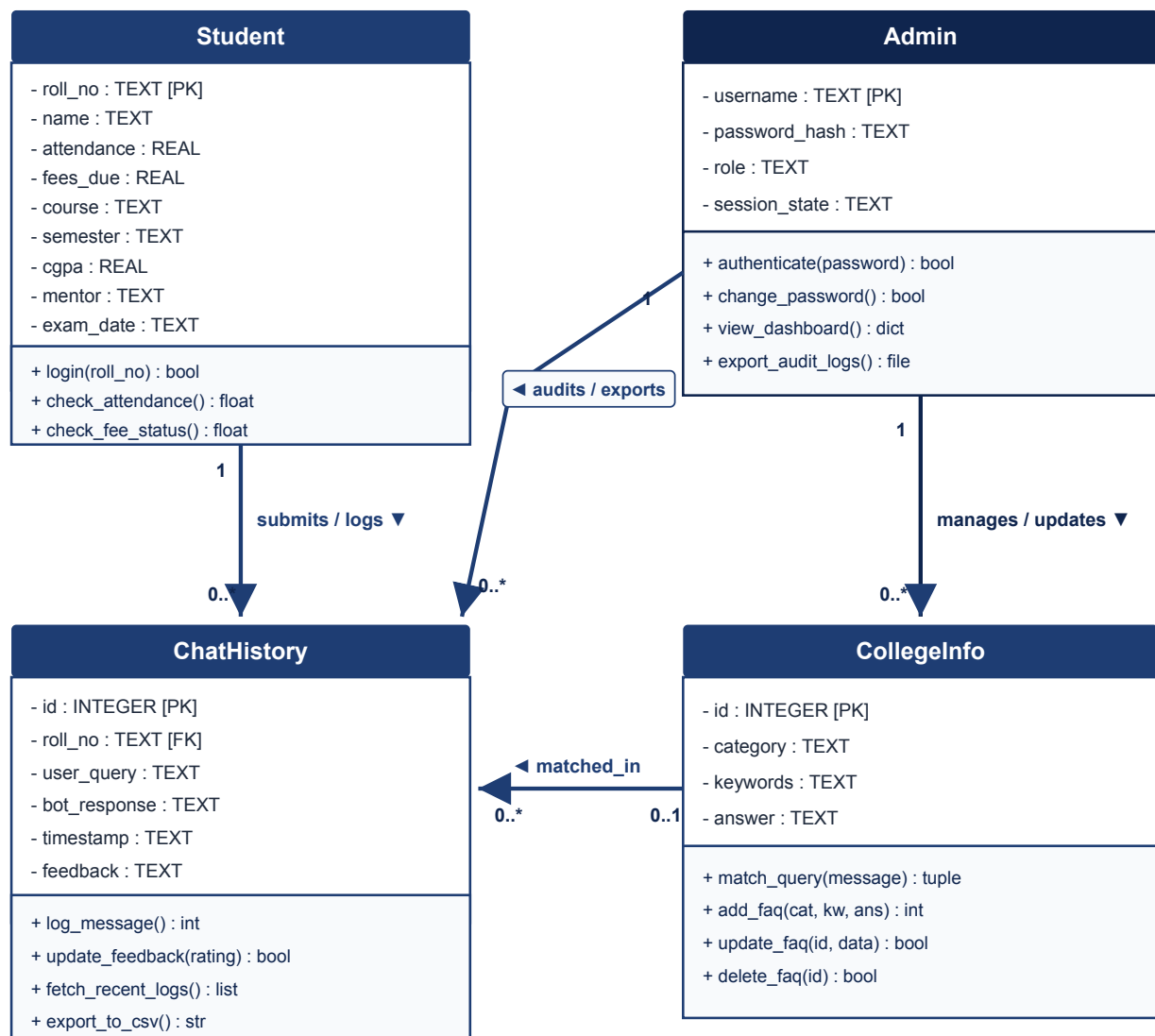


Fig 2: Class Diagram — core entities, attributes, operations, and 100% solid association arrows with multiplicities and role verbs.

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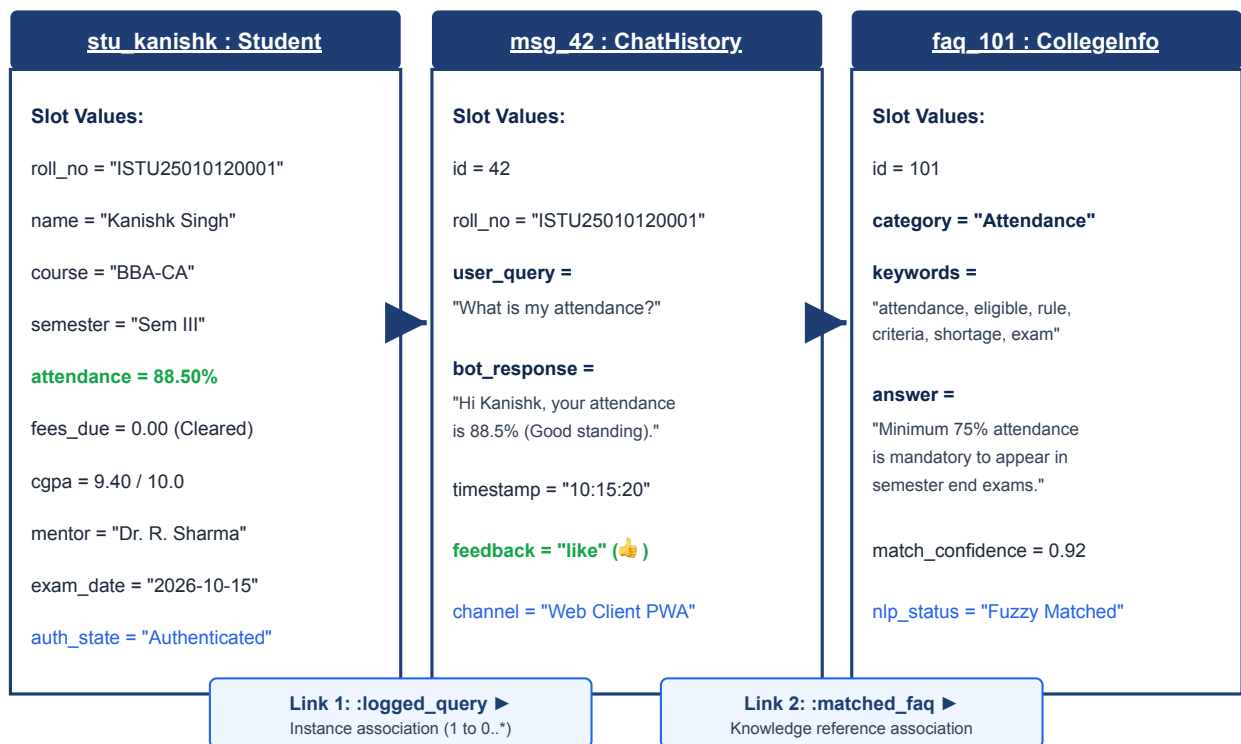
Live execution memory snapshot of an active student query transaction

3. Object Diagram (Exactly 3 Runtime Instance Boxes)

✓ Teacher Correction Applied: Strictly formatted with exactly 3 runtime object boxes (Student, ChatHistory, CollegeInfo) and solid link arrows.

Live Execution Snapshot Scenario (Active Student Transaction):

Student Kanishk Singh asks about personal attendance; ChatHistory logs query #42 matched against CollegeInfo FAQ #101.



✓ Standard UML Object Diagram Specifications Verified:

- Exactly 3 Concrete Runtime Object Instances: <u>Student</u>, <u>ChatHistory</u>, and <u>CollegeInfo</u>.
- Fully populated slot values with actual live strings, numeric metrics (88.5%, CGPA 9.40), and timestamps.
- 100% Solid Directional Association Arrows between instances (zero dashed lines).

Fig 3: Object Diagram — exactly 3 runtime instances (Student, ChatHistory, CollegeInfo) with concrete slots and solid association arrows.

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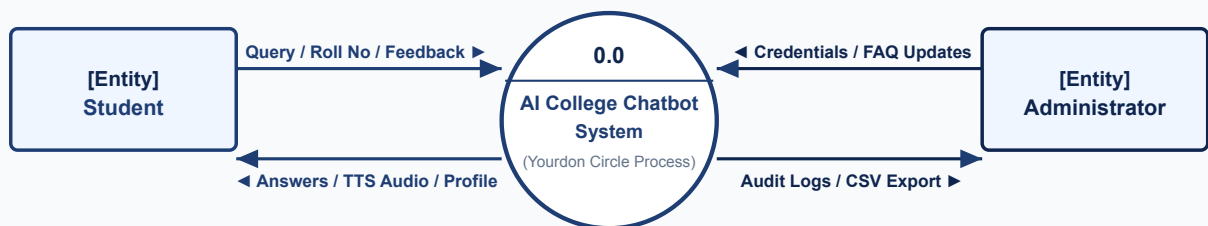
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Information flow paths across Level 0 Context and Level 1 Detailed Decomposition

4. Data Flow Diagram — DFD (Circle Symbols & Solid Arrows)

✓ Teacher Correction Applied: Adopted Yourdon & DeMarco notation using CIRCLE symbols for all processes and solid data flow arrows.

DFD Level 0: Context Diagram



DFD Level 1: Process Decomposition (Circles)

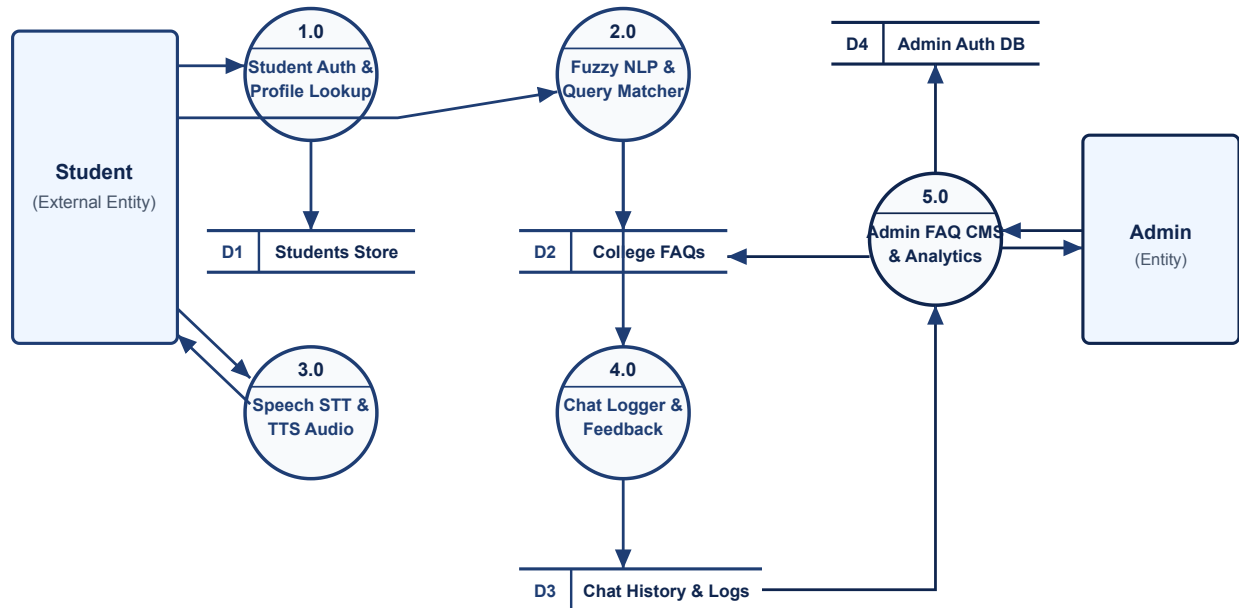


Fig 4: Data Flow Diagram (DFD) — Level 0 Context & Level 1 Process Breakdown using circle process symbols and solid data flow arrows.

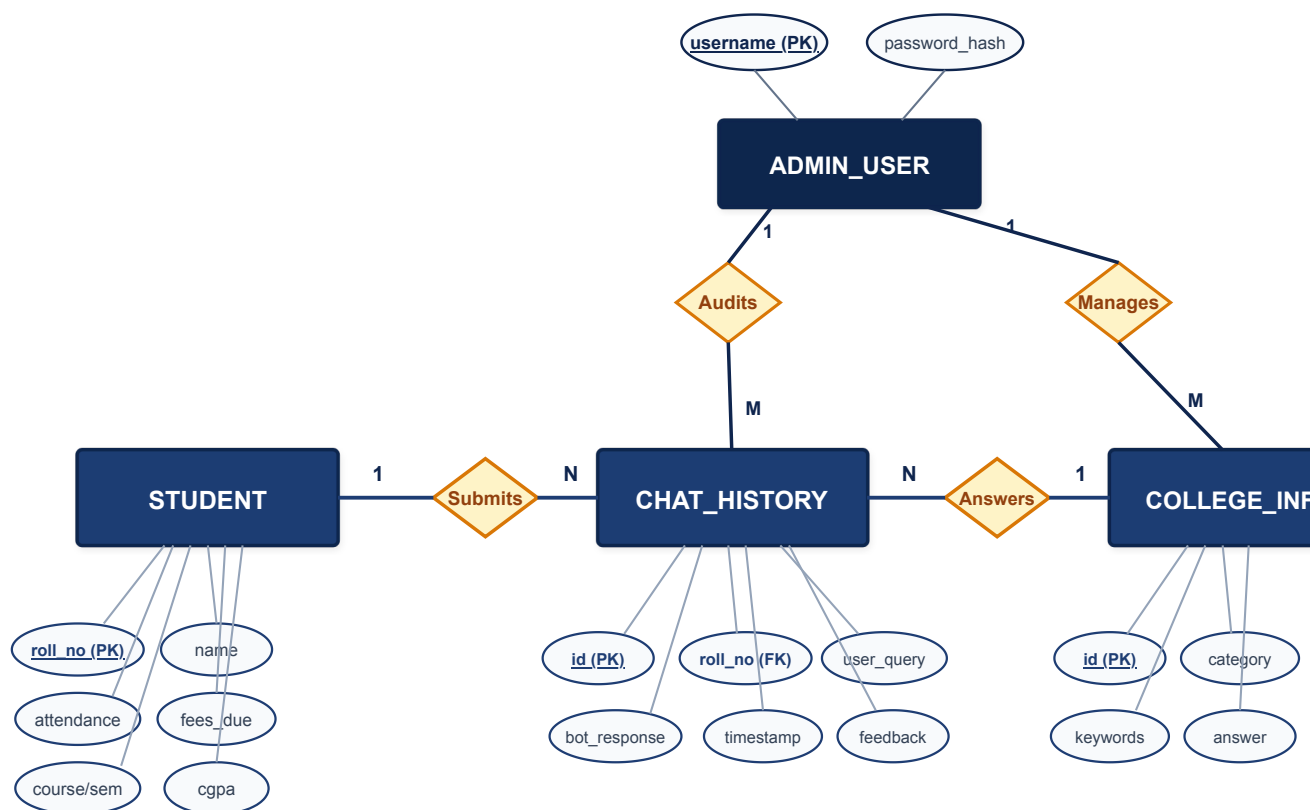
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Relational data model, primary keys, cardinalities, and non-crossing attributes

5. Entity-Relationship Diagram — ERD (Structured Relational Flow)

✓ Teacher Correction Applied: Clean left-to-right & top-to-bottom relational hierarchy with explicit relationship diamonds, cardinalities, and zero crossed lines.



✓ Standard Entity-Relationship Diagram Flow Summary:

- **Left-to-Right Primary Data Flow:** STUDENT [1] submits [N] CHAT_HISTORY, which answers [N:1] from COLLEGE_INFO.
- **Top-to-Bottom Admin Control Flow:** ADMIN_USER [1] audits [M] CHAT_HISTORY and manages [M] COLLEGE_INFO FAQs.
- **Relational Integrity:** Primary Keys underlined (<u>roll_no</u>, <u>id</u>, <u>username</u>); Foreign Key marked (roll_no FK).
- Zero line crossings; clean orthogonal layout; explicit relationship diamonds with standard cardinality markers.

Fig 5: Entity-Relationship Diagram (ERD) — structured left-to-right relational flow with non-crossing attribute branches, primary keys, and cardinalities.

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Hardware nodes, execution environments, deployed artifacts, and communication paths

6. Deployment Diagram (Textbook 3-Tier 3D UML Architecture)

✓ Teacher Correction Applied: Engineered classic 3-Tier UML 2.0 deployment model with 3D nodes (Client, App Server, DB Server) and formal protocol paths.

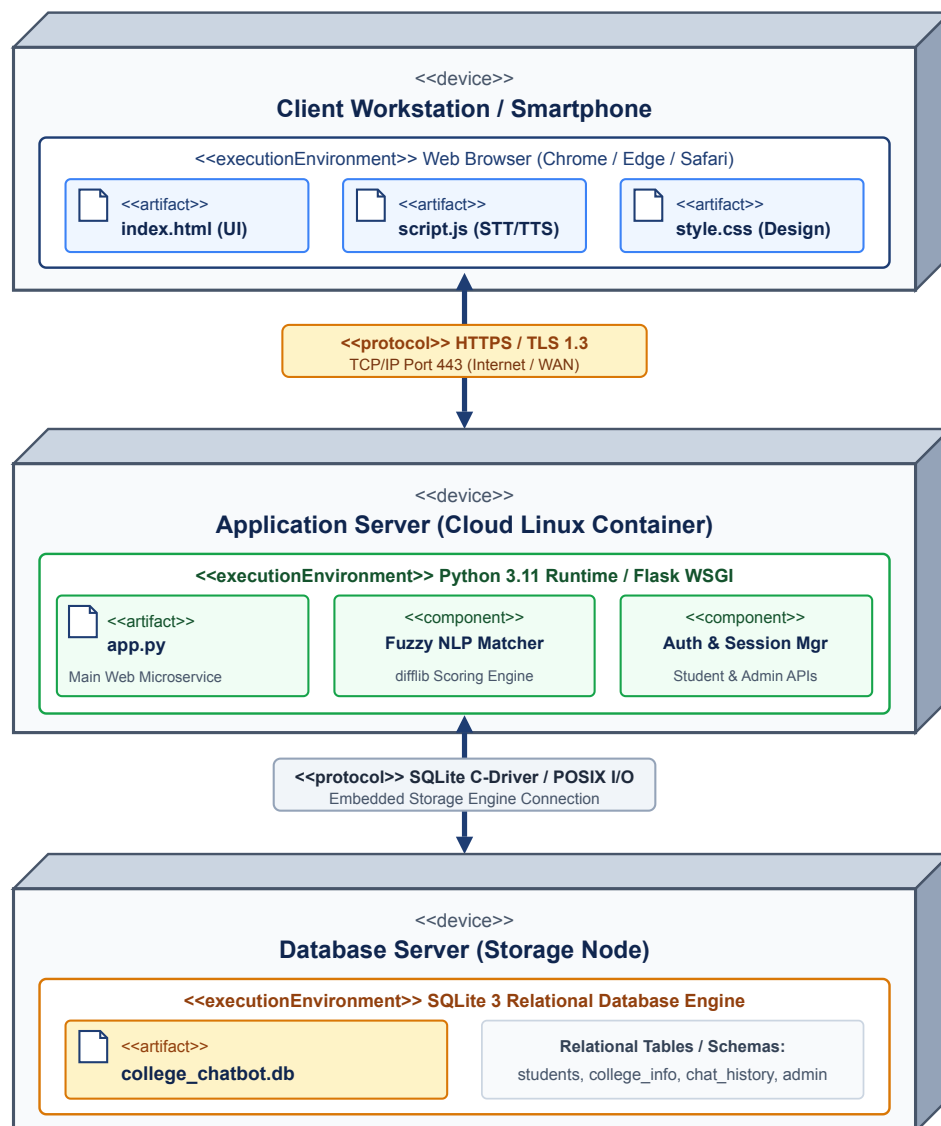


Fig 6: Deployment Diagram — textbook 3-Tier UML deployment architecture with 3D nodes (Client, Application Server, Database Server), execution environments, artifacts, and communication paths.